Selected Input Functions

- 01h, 06h - Read character from standard input
- 0Ah - Read array of buffered characters from standard input
- 0Bh - Get status of the standard input buffer
- 3Fh - Read from file or device

INT 21h Function 01h:
Read single character from standard input

- Echoes the input character
- Waits for input if the buffer is empty
- Checks for Ctrl-Break (^C)
- Acts on control codes such as horizontal Tab

```
.data
char BYTE ?
.code
mov ah,01h
int 21h
mov char,al
```

INT 21h Function 06h:
Read character from standard input without waiting

- Does not echo the input character
- Does not wait for input (use the Zero flag to check for an input character)
- Example: repeats loop until a character is pressed.

```
.data
char BYTE ?
.code
L1: mov  ah,06h          ; keyboard input
    mov  dl,0FFh         ; don't wait for input
    int  21h
    jz   L1              ; no character? repeat loop
    mov  char,al         ; character pressed: save it
    call DumpRegs        ; display registers
```

INT 21h Function 0Ah:
Read buffered array from standard input (1 of 2)

- Requires a predefined structure to be set up that describes the maximum input size and holds the input characters.
- Example:

```
count = 80

KEYBOARD STRUCT
   maxInput BYTE count          ; max chars to input
   inputCount BYTE ?            ; actual input count
   buffer BYTE count DUP(?)     ; holds input chars
KEYBOARD ENDS
```

INT 21h Function 0Ah (2 of 2)

Executing the interrupt:

```
.data
kybdData KEYBOARD <>

.code
    mov ah,0Ah
    mov dx,OFFSET kybdData
    int 21h
```

INT 21h Function 0Bh:
Get status of standard input buffer

- Can be interrupted by Ctrl-Break (^C)
- Example: loop until a key is pressed. Save the key in a variable:

```
L1: mov ah,0Bh        ; get buffer status
    int 21h
    cmp al,0          ; buffer empty?
    je  L1            ; yes: loop again
    mov ah,1          ; no: input the key
    int 21h
    mov char,al       ; and save it
```

Example: String Encryption

Reads from standard input, encrypts each byte, writes to standard output.

```
XORVAL = 239                ; any value between 0-255
.code
main PROC
     mov  ax,@data
     mov  ds,ax
L1:  mov  ah,6              ; direct console input
     mov  dl,0FFh           ; don't wait for character
     int  21h               ; AL = character
     jz   L2                ; quit if ZF = 1 (EOF)
     xor  al,XORVAL
     mov  ah,6              ; write to output
     mov  dl,al
     int  21h
     jmp  L1                ; repeat the loop
L2:  exit
```

INT 21h Function 3Fh:
Read from file or device

- Reads a block of bytes.
- Can be interrupted by Ctrl-Break (^C)
- Example: Read string from keyboard:

```
.data
inputBuffer BYTE 127 dup(0)
bytesRead WORD ?
.code
mov  ah,3Fh
mov  bx,0                       ; keyboard handle
mov  cx,127                     ; max bytes to read
mov  dx,OFFSET inputBuffer      ; target location
int  21h
mov  bytesRead,ax               ; save character count
```